HR Analytics Dashboard

Project Report

Power BI • DAX • Power Query • HR Analytics

Analyze • Visualize • Decide

Data Analytics Portfolio Project

1. Project Overview

Attribute	Details
Project Title	HR Analytics Dashboard Using Power BI
Tool	Power BI Desktop
Data Source	Embedded HR Dataset (no external connection required)
Report Pages	Single-page interactive dashboard
Domain	Human Resources / People Analytics
Purpose	Attrition analysis, workforce distribution, job satisfaction

2. Objective

The objective of this project is to analyze organizational HR data to derive meaningful insights related to employee attrition, workforce distribution, and job satisfaction. This dashboard supports data-driven HR decision-making by identifying patterns in employee behavior, retention, and demographic structure.

3. Problem Statement

Organizations often face challenges in understanding the following HR-related questions:

- Why do employees leave the organization?
- Which departments have higher attrition rates?
- How do demographics influence employee retention?
- How does job satisfaction vary across roles?

This project addresses these challenges by visualizing HR data in an interactive, filter-driven Power BI dashboard.

4. Dataset Description

The dataset used in this project is an embedded HR dataset within Power BI. No external database connection is required — all data is packaged within the .pbix file.

4.1 Key Fields

Field	Description
Employee Count	Total headcount in the organization
Job Satisfaction	Satisfaction rating (scale 1–4)
Attrition Count	Number of employees who left
Attrition Rate	Percentage of employees who left
Active Employees	Current active workforce size
Age	Employee age (continuous)
Age Band	Grouped age ranges (e.g., 25–34, 35–44)
Gender	Employee gender
Department	Organizational department
Job Role	Specific role or designation
Education Field	Field of academic background

5. Tools & Technologies

Tool / Technology	Purpose
Power BI Desktop	Dashboard design, layout, and report publishing
Power Query (M Language)	Data ingestion, cleaning, and transformation
DAX (Data Analysis Expressions)	Calculated measures — Attrition Rate, KPIs
Data Visualization Techniques	Charts, slicers, KPI cards, matrix tables

6. Data Preparation & Transformation

The following preprocessing steps were performed using Power Query to ensure clean and structured data for visualization:

- Removal of null or inconsistent values
- Creation of calculated fields (e.g., Attrition Rate = Attrition Count / Employee Count)
- Age grouping into defined Age Bands for demographic analysis
- Data type standardization across all columns
- Categorization of Education Field and Job Roles for consistent grouping

7. Data Modeling Approach

A flat data model structure was used since the dataset is fully embedded within Power BI. Key relationships were maintained implicitly within Power BI visuals, eliminating the need for a complex star or snowflake schema.

Modeling Decision	Rationale
Flat model (single table)	Embedded dataset with no relational joins required
Implicit relationships	Power BI visuals handle context filtering internally
Calculated columns	Age Band and categorized fields added via Power Query
DAX measures	Attrition Rate and KPIs defined as explicit measures

8. Dashboard Overview

The dashboard is designed as a single-page interactive report. All visuals are cross-filtered and connected through a shared slicer for Education Field.

8.1 KPI Cards

KPI Card	Metric Displayed
Total Employee Count	Overall headcount of the organization
Attrition Count	Total number of employees who left
Attrition Rate	Percentage of workforce lost
Active Employees	Current active headcount
Average Age	Mean age across the workforce

8.2 Visual Analytics

Visual Type	Fields Used	Purpose
Pie Chart	Department, Attrition Count	Attrition share by department
Bar Chart	Education Field, Attrition Count	Attrition count by education background
Column Chart	Age Band, Gender, Employee Count	Workforce distribution by age and gender
Donut Charts	Gender, Age Band, Attrition Count	Attrition breakdown by gender and age group
Matrix Table	Job Role, Job Satisfaction, Count	Satisfaction scores across all job roles

8.3 Filters & Interactivity

- Education Field slicer — dynamically filters all visuals on the page
- Cross-filtering — clicking any chart segment filters all other visuals
- Hover tooltips — provide granular values on every data point

9. Key Insights

9.1 Attrition Patterns

Certain departments show significantly higher attrition rates compared to others, indicating potential organizational or managerial challenges within those teams.

9.2 Demographic Influence

Younger age groups tend to exhibit higher attrition rates, suggesting early-career mobility and a preference for job-switching in the initial years of employment.

9.3 Education Impact

Employees from specific education backgrounds show varying levels of retention and job stability. Certain fields are associated with higher turnover.

9.4 Job Satisfaction Trends

Job satisfaction varies significantly across job roles. Lower satisfaction scores in certain roles directly correlate with higher attrition within those positions.

9.5 Workforce Composition

The organization maintains a broadly balanced distribution of employees across gender and age groups, with minor demographic variations visible in the column chart.

10. Business Impact

This dashboard enables HR teams to make faster, evidence-based decisions:

- Identify risk areas for employee turnover before they escalate
- Improve retention strategies based on demographic and satisfaction data
- Optimize workforce planning using headcount and age distribution insights
- Monitor employee satisfaction trends across job roles over time
- Support strategic HR decisions with a single source of visual truth

11. Conclusion

The HR Analytics Dashboard successfully transforms raw HR data into actionable insights. It provides a clear understanding of workforce behavior, attrition trends, and employee satisfaction levels across departments and demographics.

Such dashboards are essential for modern HR analytics — enabling organizations to move away from reactive decision-making and toward proactive, data-driven HR strategy.

12.Future Enhancements

- Predictive attrition modeling using Machine Learning (Python / Azure ML)
- Time-series analysis of employee trends across quarters and years
- Drill-through pages for department-level and role-level deep analysis
- Integration with live HR databases (SAP, Workday, BambooHR)
- Role-based access dashboards for senior management vs. HR teams

Built as part of a Data Analytics Portfolio • Power BI • DAX • Power Query